Inferential statistics

Using sample data to draw conclusions about a population

Inferential statistics this week and next

1. Estimation
2. Significance tests
3. Linear models
4. Multiple testing

Estimating population parameters

Weddell seal dive durations



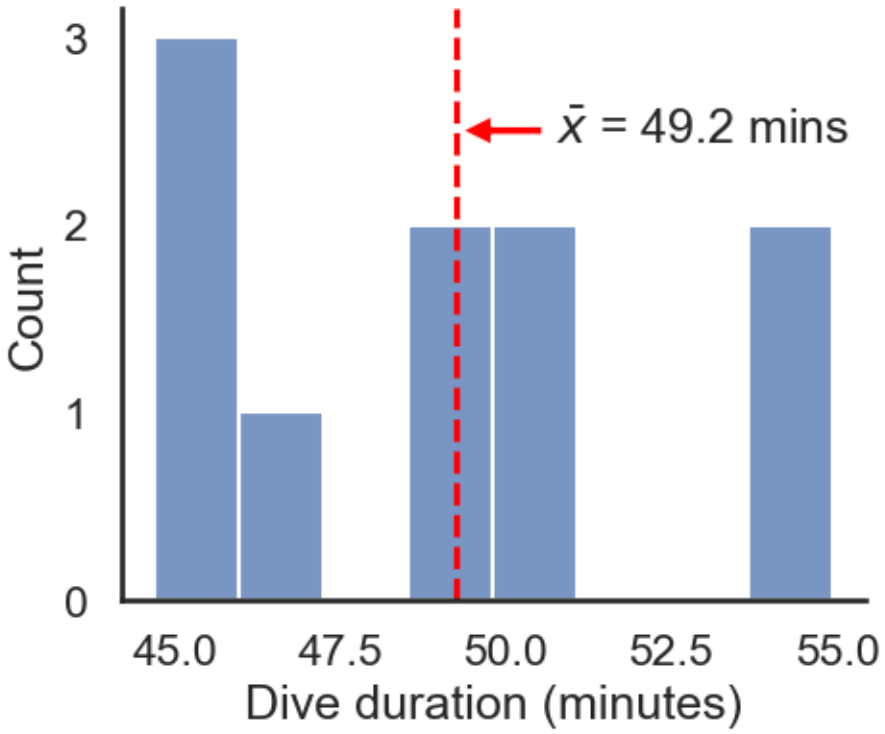
- Marine biologists want to estimate the mean dive duration of weddell seals
- Estimated population size in Antarctica ~200,000 seals
- They cannot measure every dive of every seal in Antarctica
- They therefore measured dive duration in a sample of 10 seals from the Ross sea by attaching satellite telemetric tags

Dive duration (mins)
49.5
46.3
50.2
54.6
45.8

Dive duration (mins)
45.8
54.9
50.8
44.7
49.7
$\bar{x} = \mathbf{49.2}$

The sample mean is denoted $\bar{x}$

Distribution of dive durations



- The mean dive duration of this sample is 49.2 minutes
- But we would like to know the overall mean duration of all Weddell seal dives
- This is called the **population mean**, usually denoted μ
- But we can never know the population mean because we can't measure every single dive
- The best we can do is estimate it
- A **point estimate** of the population mean is usually denoted $\hat{\mu}$

- The best point estimate of the population mean is the sample mean:

$$\hat{\mu} = \bar{x} = 49.2\,\text{mins}$$

A point estimate is not enough information

But what if the marine biologists sampled a different set of 10 weddell seal dives

Sample 1	Sample 2	Sample 3
49.5	44.7	45.7
46.3	49.3	43.4
50.2	48.2	47.3
54.6	37.4	39.9
45.8	38.4	44.3
45.8	44.2	44.3
54.9	41.9	46.5
50.8	48.6	43.2
44.7	42.5	45.0
49.7	54.3	44.0
$\bar{x}_1 = \mathbf{49.2}$	$\bar{x}_2 = \mathbf{45.0}$	$\bar{x}_3 = \mathbf{44.4}$

Three different point estimates

- We have just one point estimate of the population mean ($\hat{\mu} = 49.2$ minutes)
- But there is some uncertainty in its value because only 10 dives were recorded

$$\hat{\mu} = 49.2 \pm \text{uncertainty}$$

- How much uncertainty is there? For example, is it:
  - 49.2 ± 10 mins (around 39 to 59 mins)?
  - 49.2 ± 1 mins (around 48 to 50 mins)?
- We must quantify the uncertainty in dive durations

The uncertainty in the estimate of a population mean

There are two common methods of reporting the uncertainty in the estimate of a population mean

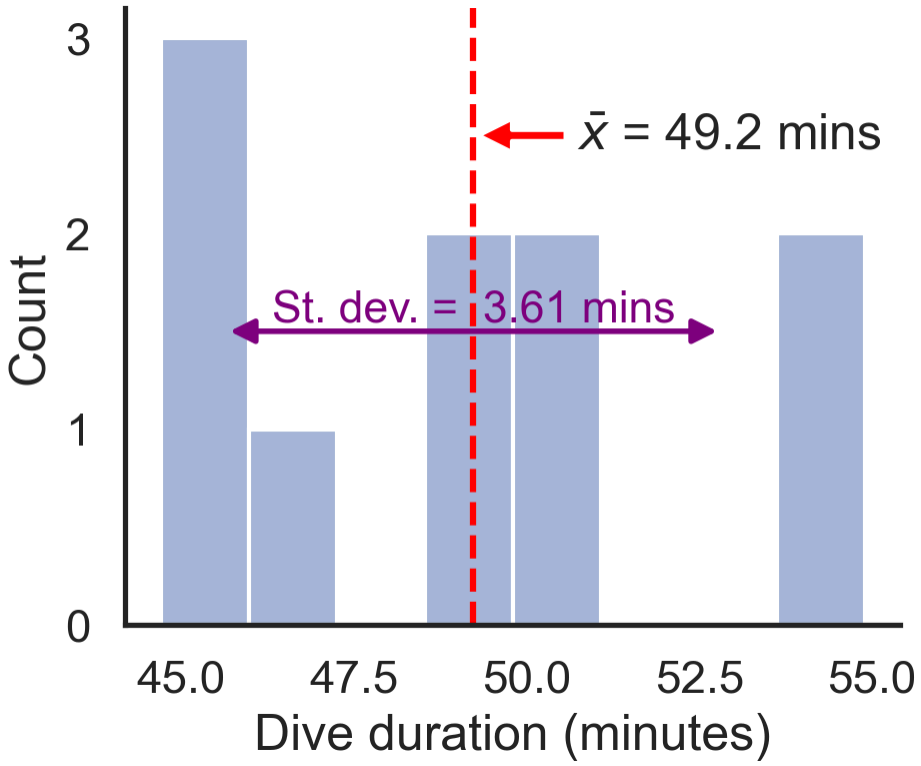
1. Standard error of the mean (SEM)
   - Good when your sample size is more than about 20
   - But is inaccurate when the sample size is less than about 20
2. 95% confidence interval
   - Good for all sample sizes
   - SEM is used to calculate 95% CI

The standard error of the mean

The formula for the standard error of the mean is

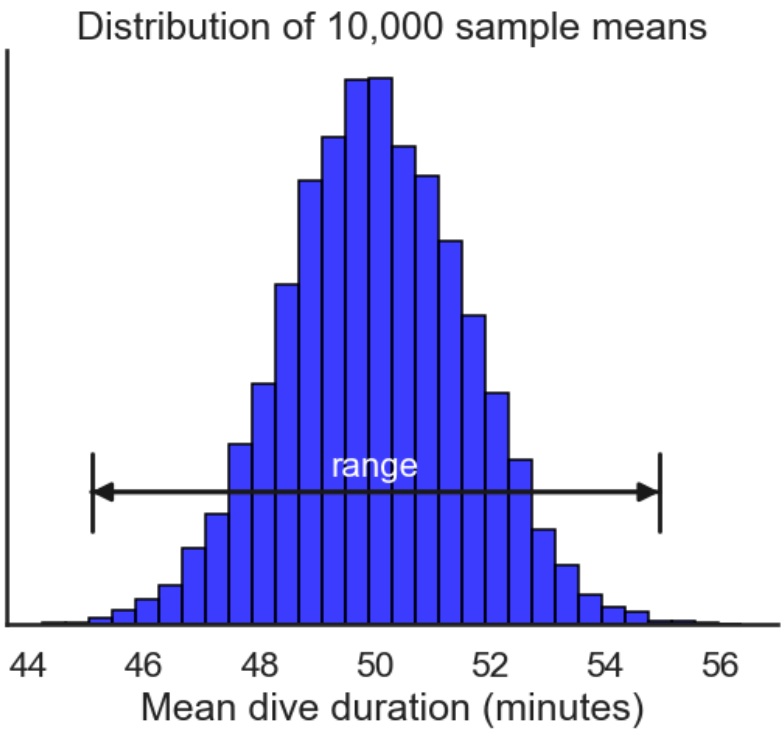
$$\text{SEM} = \frac{\text{Standard deviation of the sample}}{\text{Square root of number of observations}} = \frac{s}{\sqrt{n}}$$

Where does this formula come from?

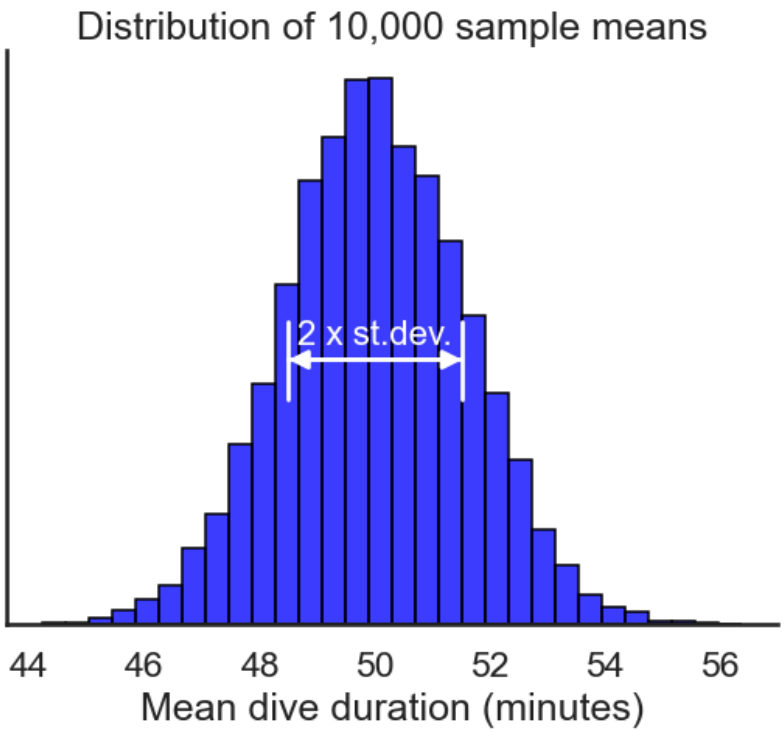


The sampling distribution

- Say you had 10,000 samples, each of 10 dive durations
- Each sample has a mean. These will differ between samples
- The distribution of these 10,000 means is called a sampling distribution



- So samples of 10 dive durations have means roughly in the range 45 to 55 minutes
- That's a range of about ± 5 minutes
- The width of the sampling distribution tells us the uncertainty in an estimate of the population mean
- The mean of the sample measured by the marine biologists will lie somewhere in this sampling distribution
- So the uncertainty in the population mean estimate of 49.2 minutes is about ± 5 minutes



- Actually we don't use the **range** of the sampling distribution, instead we use its **standard deviation**
- The standard deviation of the sampling distribution is called the **standard error of the mean**

It turns out, and you can prove this mathematically, that SEM is related to:

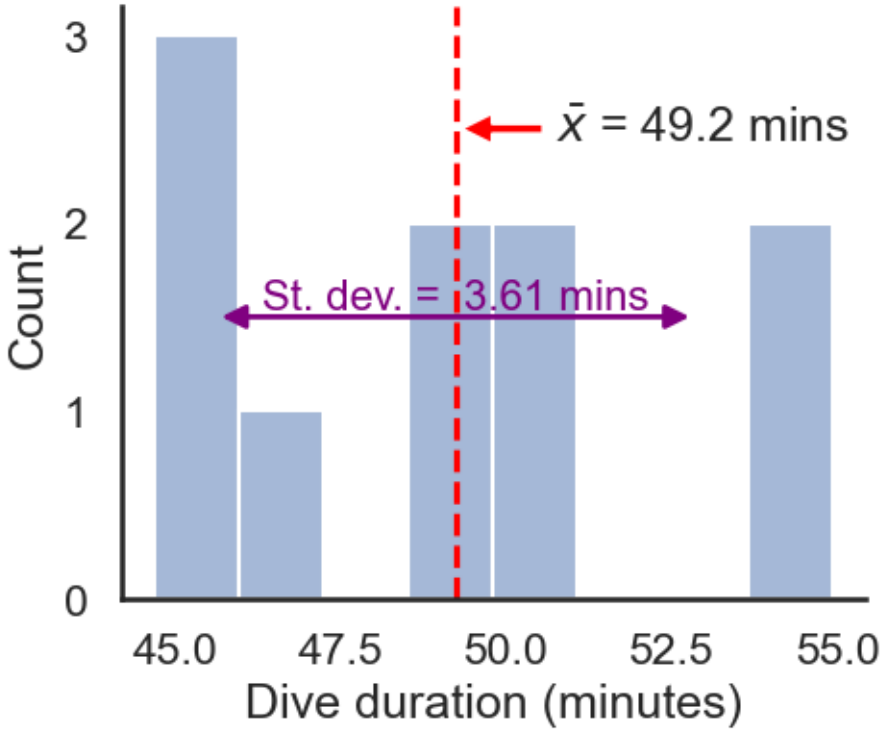
1. the variability of all dive durations (the population standard deviation, σ) and
2. the sample size, n

$$\text{SEM} = \frac{\text{Standard deviation of the population}}{\text{Square root of number of observations}} = \frac{\sigma}{\sqrt{n}}$$

- However, we don't know the standard deviation of the population, σ
- But we do know the standard deviation of the sample, s
- s is an estimate of σ
- So replace σ with s

$$\text{SEM} = \frac{\text{Standard deviation of the sample}}{\text{Square root of number of observations}} = \frac{s}{\sqrt{n}}$$

Calculating SEM



Dive duration (mins)
49.5
46.3
50.2
54.6
45.8
45.8
54.9
50.8
44.7
49.7
$\bar{x} =$ **49.2**

```
dives = pd.read_csv('../Datasets/weddell_seal_dives.csv')

n = dives['dive_duration_min'].count()
```

```
xbar = dives['dive_duration_min'].mean()
s = dives['dive_duration_min'].std()

# Either
sem = s/np.sqrt(n)
# or
sem = dives['dive_duration_min'].sem()

print(f'n:    {n}')
print(f'xbar: {xbar:.2f} mins')
print(f's:    {s:.2f} mins')
print(f'SEM:  {sem:.2f} mins')
```

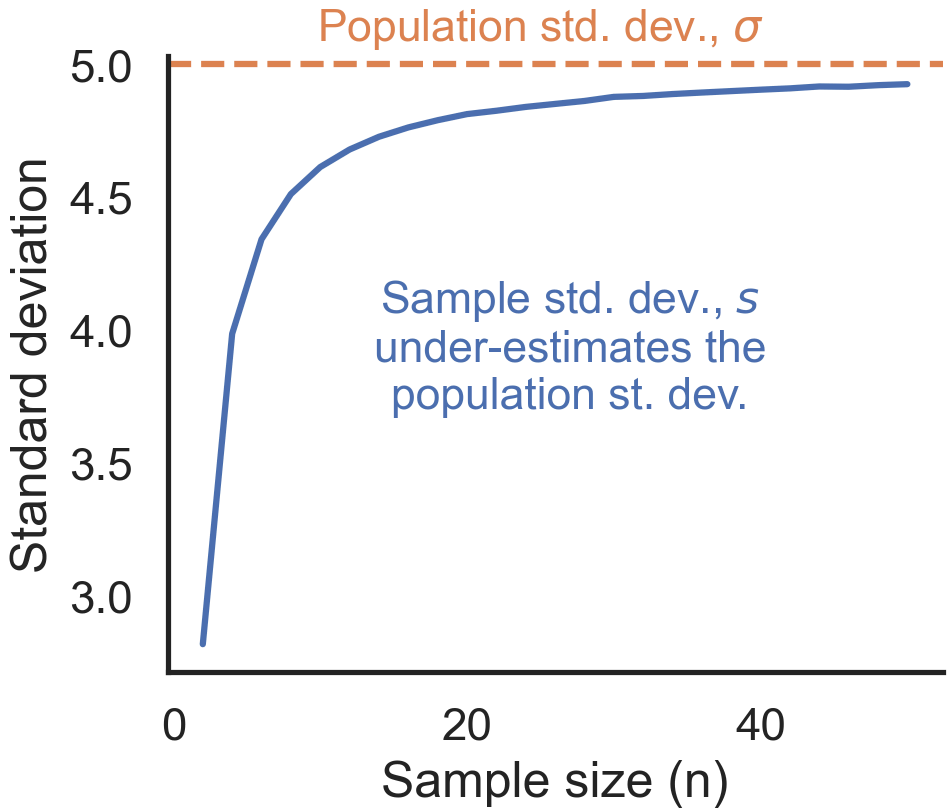
```
n:    10
xbar: 49.23 mins
s:    3.61 mins
SEM:  1.14 mins
```

The 95% confidence interval

To calculate SEM we replaced the standard deviation of the population σ (which we don't know) with the standard deviation of the sample s (which we do know)

$$\text{SEM} = \frac{\text{Standard deviation of the sample}}{\text{Square root of number of observations}} = \frac{s}{\sqrt{n}}$$

- This assumes that s is a good estimate of σ
- But when the sample size is small, s is *not* a good estimate of σ
- The sample standard deviation s underestimates the population standard deviation σ.
- And the underestimate gets worse at smaller sample sizes



So for small sample sizes we have to include a correction factor called t_{n-1}

$$\hat{\mu} = \bar{x} \pm t_{n-1} \times \text{SEM}$$

The factor t_{n-1} is a correction factor for small sample sizes.

n	t_{n-1}
2	12.71
5	2.78
10	2.26
20	2.09
50	2.01
100	1.98
500+	1.96

A rule of thumb:

For samples with more than 20 observations t_{n-1} is about 2

$$\hat{\mu} = \bar{x} \pm 2 \times \text{SEM}$$

The code to calculate t_{n-1} is

```
from scipy.stats import t

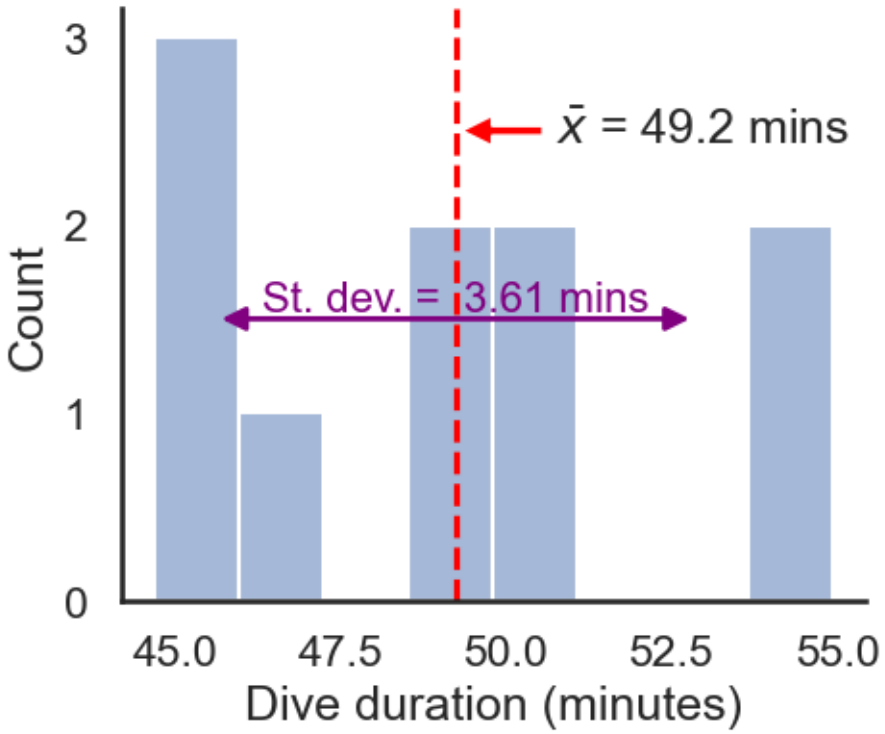
t_df = t.ppf(0.975, df=9)

print(f't_df = {t_df:.2f}')
```

```
t_df = 2.26
```

`df` stands for “degrees of freedom”. In this case it is one less than the sample size.

The final estimate



From the sample we have

$$
\begin{aligned}
n &= 10 \\
\bar{x} &= 49.2\,\text{mins} \\
\text{SEM} &= 1.14\,\text{mins} \\
t_9 &= 2.26
\end{aligned}
$$

$$\hat{\mu} = \bar{x} \pm t_{n-1} \times \text{SEM}$$
$$\hat{\mu} = 49.2 \pm 2.26 \times 1.14$$
$$\hat{\mu} = 49.2 \pm 2.6 \,\text{mins}$$

In conclusion, mean dive durations are about 49 minutes with an uncertainty of plus/minus two and a half minutes.

The 95% confidence interval

95% CI are often written as a range from the low bound to the high bound.

- low bound = $\bar{x} - t_{n-1} \times \text{SEM}$
- high bound = $\bar{x} + t_{n-1} \times \text{SEM}$

```
ci_low  = xbar - t_df * sem
ci_high = xbar + t_df * sem

print(f'95% CI: [{ci_low:.1f}, {ci_high:.1f}] mins')
```

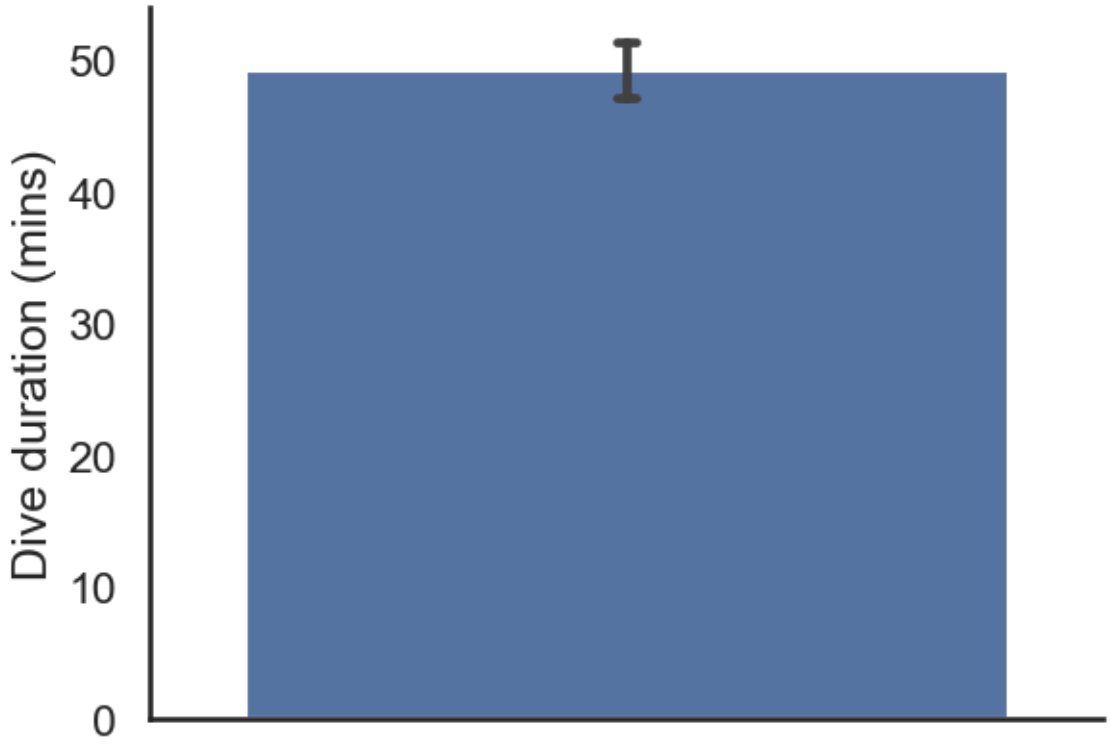
```
95% CI: [46.6, 51.8] mins
```

- The 95% CI of dive durations is [46.6, 51.8] mins.

Visualising estimates

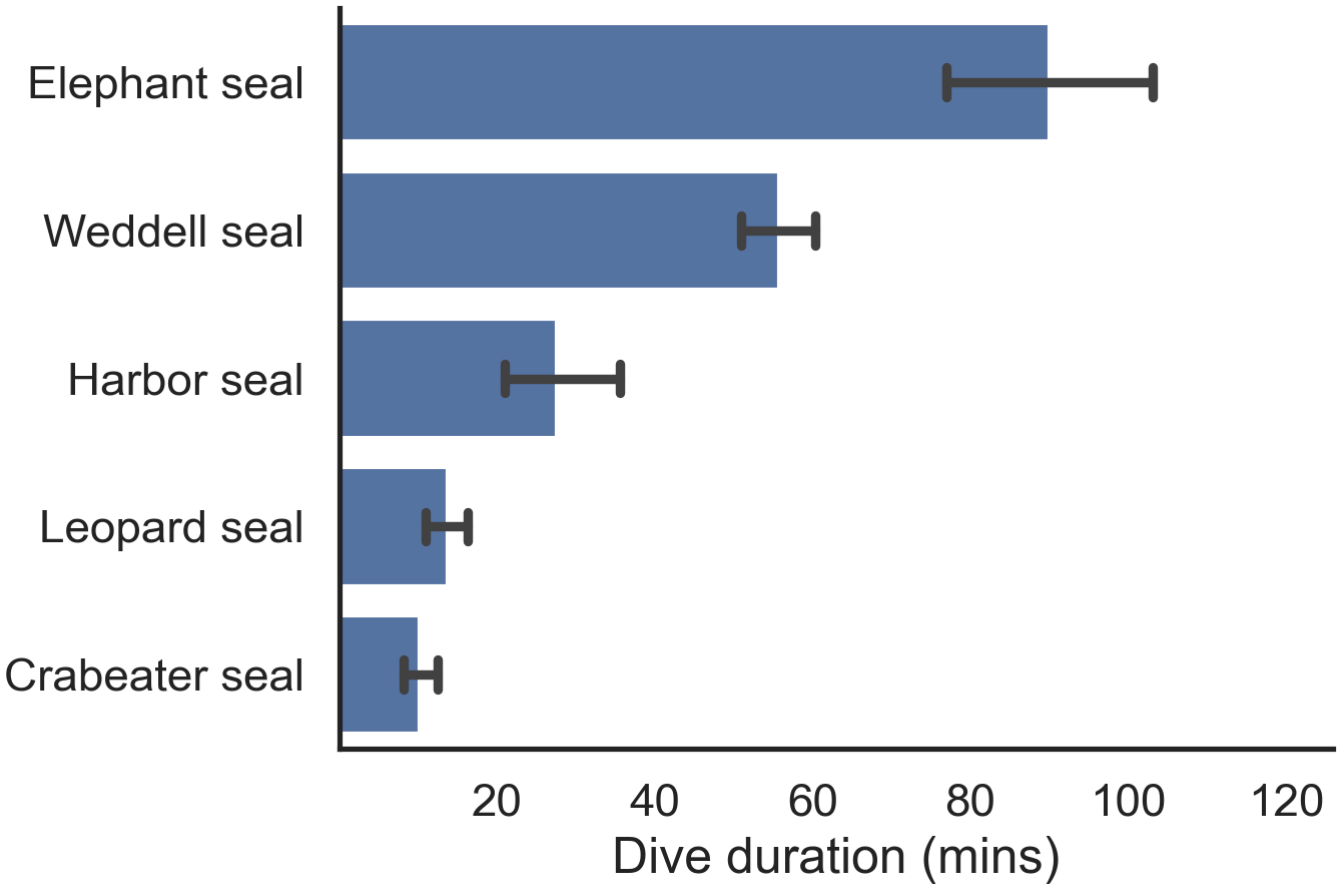
- Uncertainty in point estimates are graphically represented as error bars
- Barplots are a common method

```
ax = sns.barplot(data=dives, y='dive_duration_min', capsize=.02);
ax.set_ylabel('Dive duration (mins)');
```



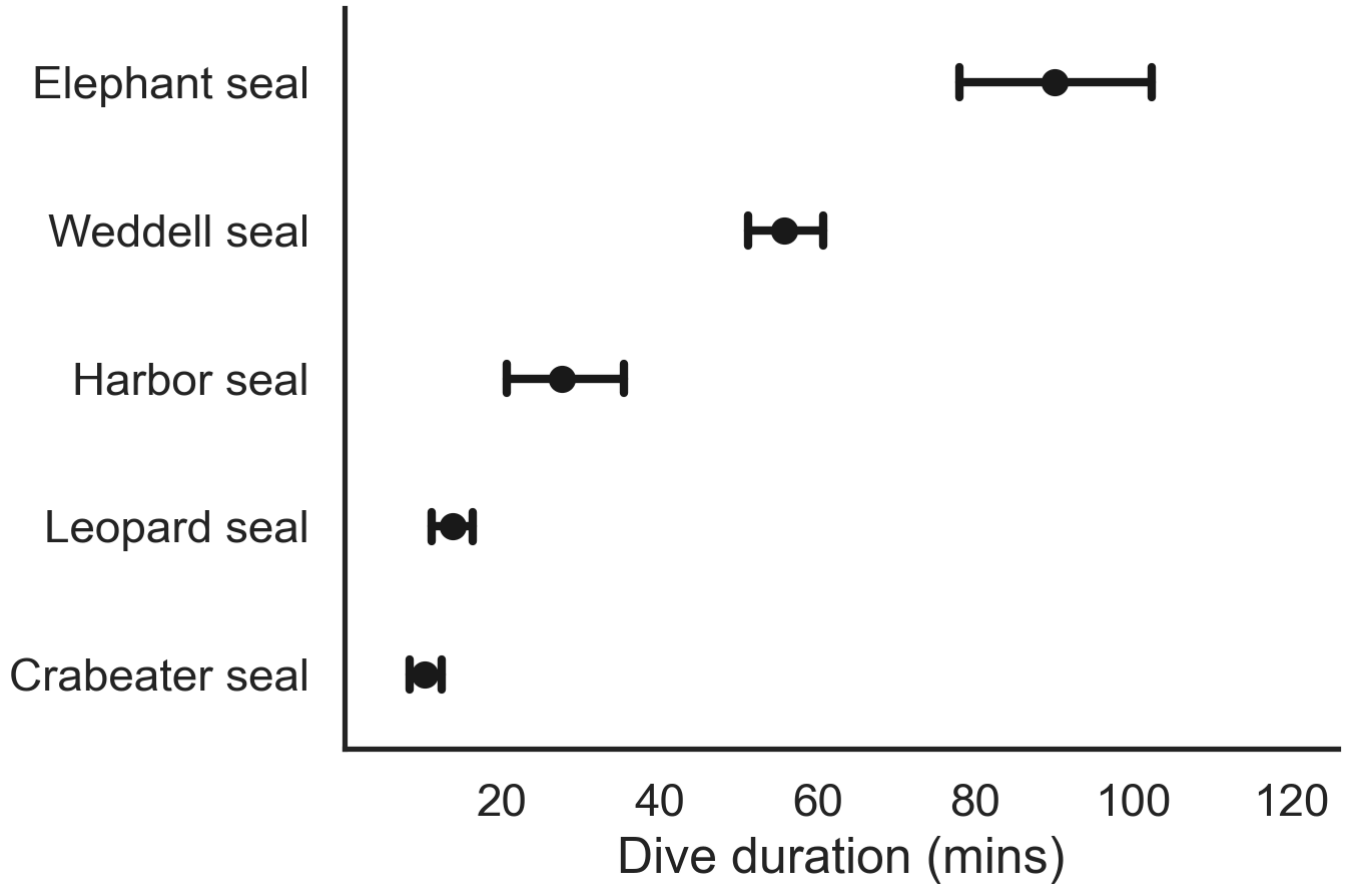
A bar plot creates a bar with the bottom at zero and the top at the point estimate. The errorbar extends from the low bound to the high bound of the 95% CI.

These plots are more useful when comparing things. Dive durations of different seal species, for example.



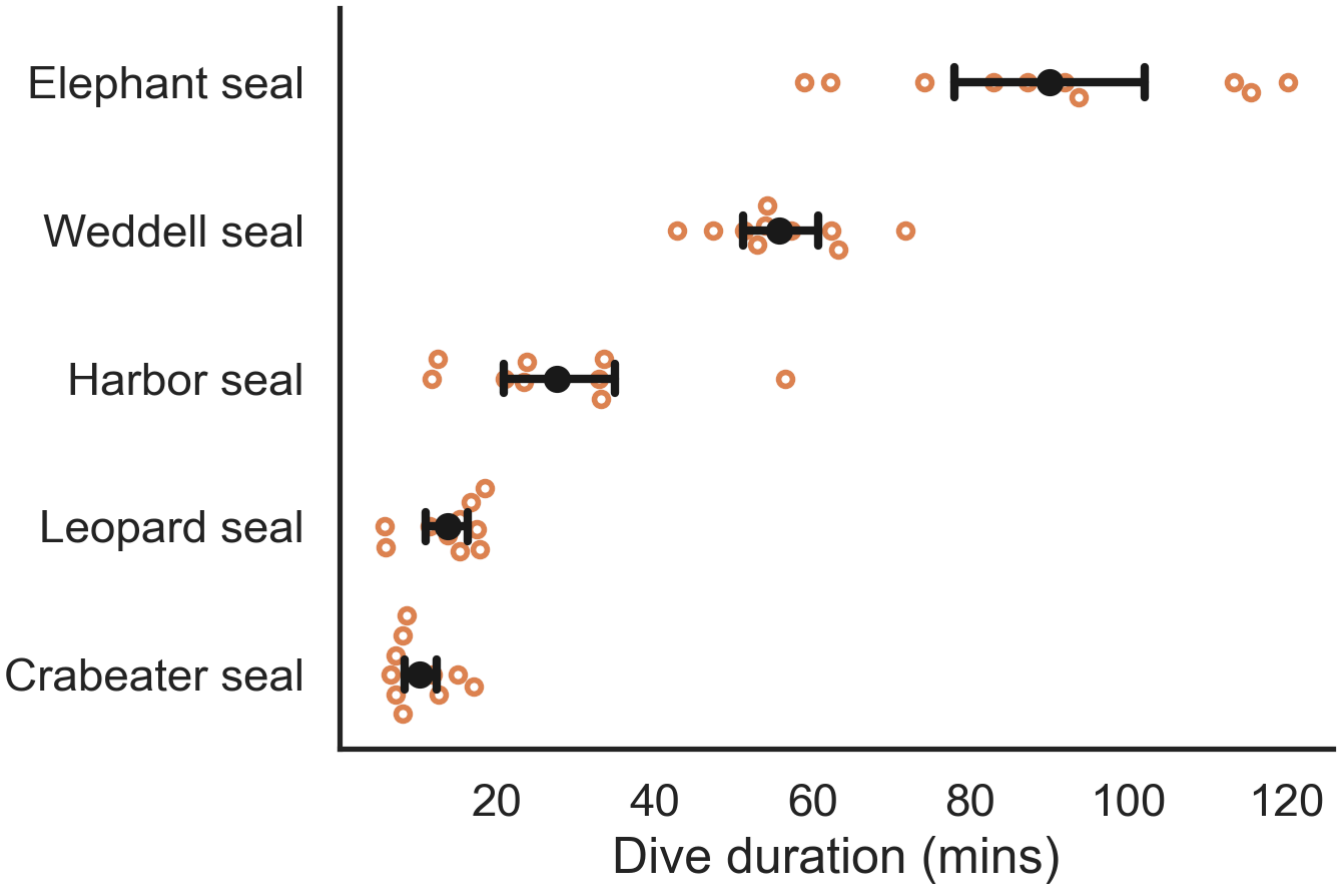
Error bars in a point plot

A point plot just plots a circle at the point estimate. This gives a less cluttered figure than a bar plot.



Include the data

If there are not too many data including them can be useful to see the distribution of the data



Always state what your error bars represent

> ! Important
>
> Always state in the figure caption what errorbars represent (SEM, 95% CI) so that readers know how to interpret them.

Conclusion

Whenever we estimate a population parameter we must always include the uncertainty in that estimate.

This week's class notebook

Three questions:

1. **Weddell seal dive durations**:
   - The same analysis as in the lecture but with a larger data set
2. **Evolution of ground finch beak depths**:

- Estimate the **difference** in the means of two populations
- This is a little more complicated and is a prelude to Friday's material

3. **Proportion of hospital patients with vancomycin-resistant intestinal *Enterococcus* bacteria**

    - How to estimate a population proportion from a sample of data
    - Method 1: By using a formula for the standard error of the proportion
    - Method 2: By using Monte-Carlo simulation of the sampling distribution of a proportion